

## # Internal Audit Reveals Undisclosed 2024 Telemetry Faults in Apex Sensor Firmware

**\*\*By David Rivera | Investigative Dispatch | October 22, 2024\*\***

**METRO CITY** — A confidential internal quality audit obtained by Investigative Dispatch reveals that Northstar Technologies discovered persistent calibration drift in its Apex 3.0 municipal telemetry sensors six months before alerting regional transportation authorities.

According to the 48-page audit authored by senior systems engineer Dr. Maya Lin, optical-inertial sensors deployed across thirty-four regional bridges exhibited temperature-induced calibration skew during peak summer months. The sensor error produced false-negative readings regarding structural load stress, potentially masking micro-fractures in concrete foundations.

"The telemetry firmware version 3.2.1 contained an unverified dampening algorithm that artificially smoothed out high-frequency stress spikes," Dr. Lin noted in an internal memorandum dated August 14, 2024. "Rather than flagging genuine structural vibration anomalies, the firmware misclassified them as ambient sensor noise."

When approached for comment, Northstar Technologies communications vice president Julian Vance stated that the issue was confined to early production batches and has since been resolved with firmware patch 3.2.4. However, municipal records indicate that Metro City Transit Authority has filed a formal \$12.5 million warranty arbitration claim against Northstar to recoup replacement and manual inspection costs.

State infrastructure regulators confirmed yesterday that an expedited technical review is underway, with preliminary findings expected by December 2024.